Programming and Spreadsheet Modeling

Technical overview

Some graphics and datasets used in this presentation and accompanying MS Excel models are created by artificial intelligence. While they are realistic, they are not from actual companies or pre-existing data sets.

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Programming and Spreadsheet Modeling
Course technical overview

- Management Science Approach
- Data Analytics Stack
- Mechanics vs. Modeling

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Management Science Approach

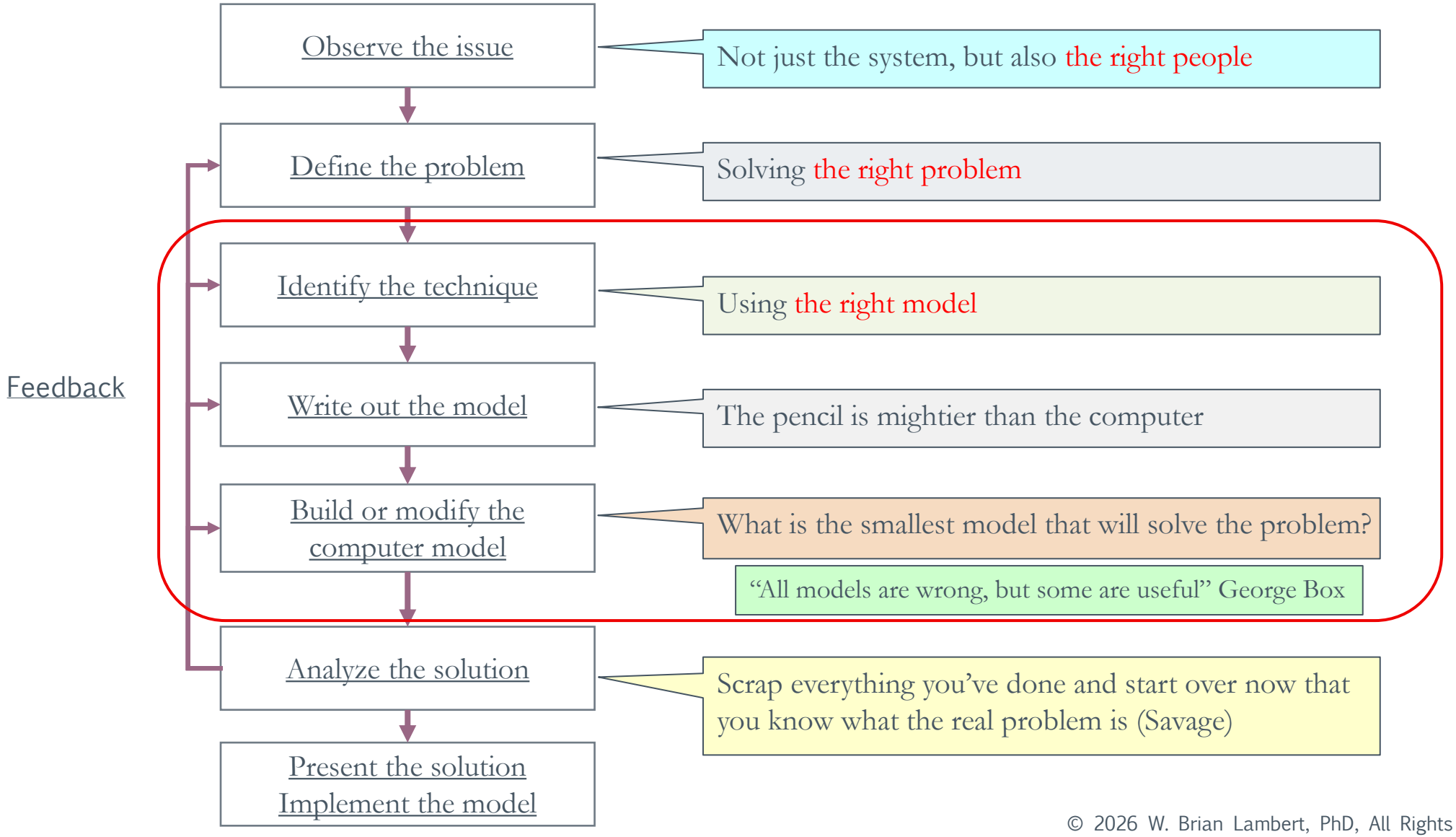
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Management Science

- Management problems are often “solved” and decisions made using human reasoning, and ad-hoc processes
- Alternatively, the scientific method can be applied to solve management problems in business
- The Management Science requires using quantitative, mathematical approaches to solving problems
- It is used across industries to solve many and varied types of problems

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The Management Science Process



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Management Science Approach
In this course

- For a given a problem:

    1. Select the appropriate technique for the problem

    2. Develop the mathematical model

    3. Implement the model in software

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Data Analytics Stack

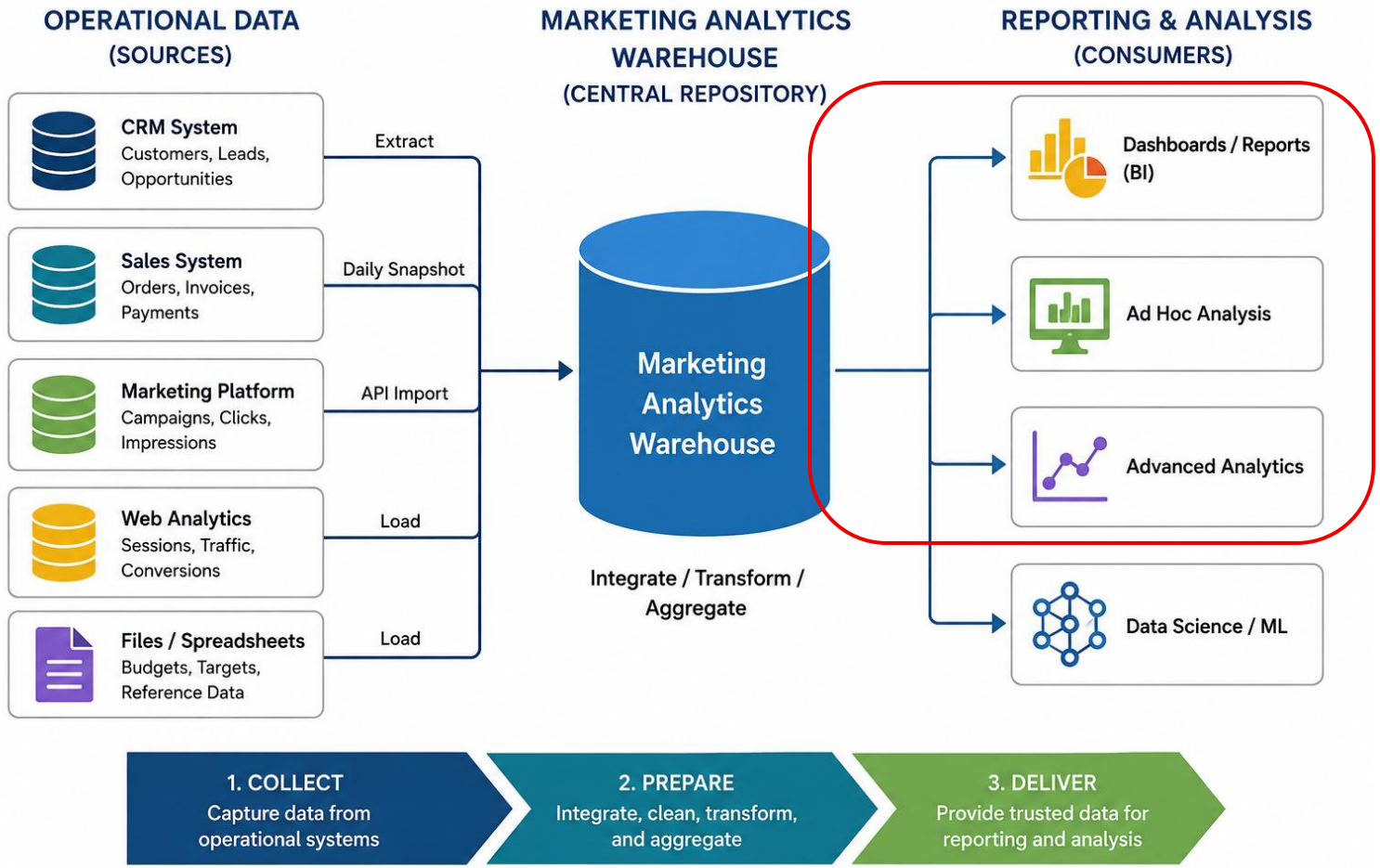
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Data Analytics Stack
Discipline domains

- Statistics – techniques and methods to analyze the data
- Data Analytics – not so much the methods but using them for some distinct objective. This requires understanding more about and working with technologies – databases, packages, APIs, etc.
- Business Analytics – data analytics used for business objectives requiring understanding of business requirements and the business environment
- Data Science – building back-end algorithms, requiring significantly deeper understanding and use of technologies

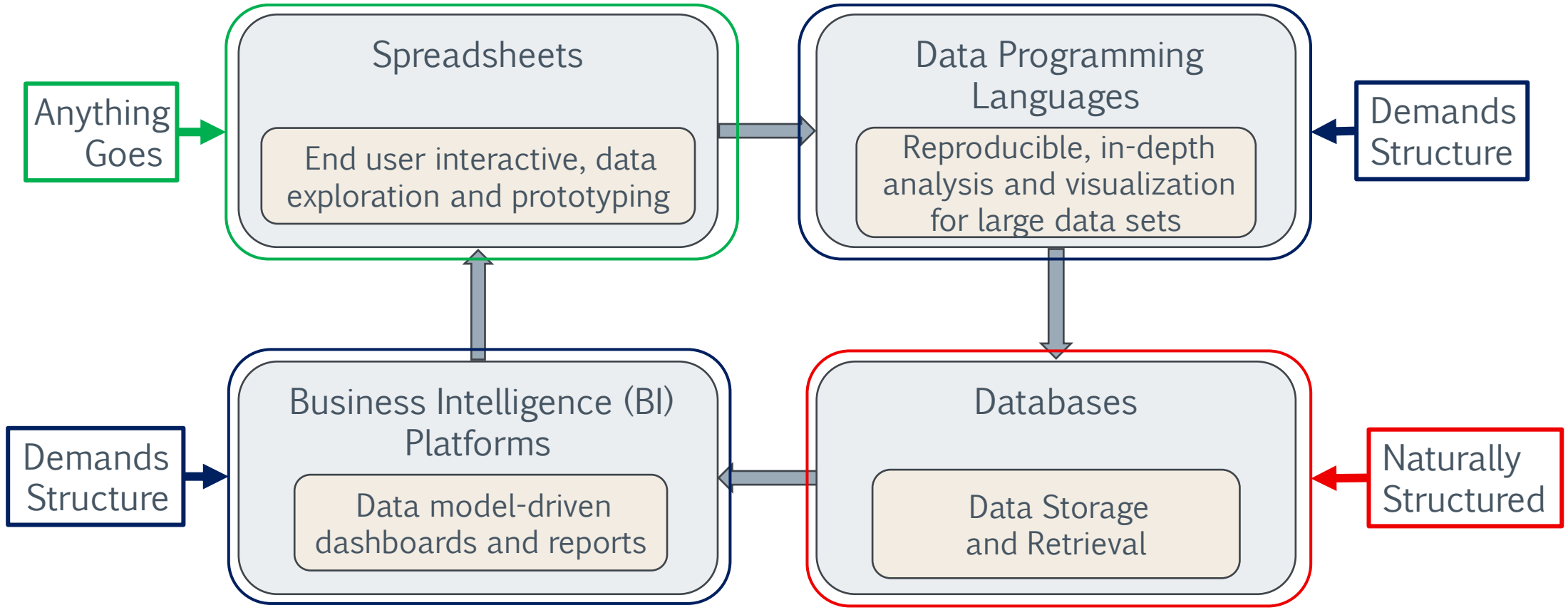
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Data Analytics Stack
In practice



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Data Analytics Stack
Multiple Tools – multiple purposes



Adapted from George Mount, *"Advancing into Analytics"*, 2021, Figure 5-1, p.82

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Mechanics vs. Modeling

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Mechanics vs. Modeling Mechanics

- The model is “what” it does while the mechanics are “how” it does it
- Mechanics are the syntax and analytical functions that we use to make the software do what we want
- Different software tools contain different functions, each with advantages and disadvantages
- Capabilities necessary for the task often exist in different tools
- This means selecting the appropriate tool may not be trivial
- A simple example: using a spreadsheet to store a moderate-sized data set may be preferable to creating a database which other users are not familiar with

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Mechanics vs. Modeling
Modeling

- Model: “what” it does
- Mechanics: “how” it does it
- The model should drive the technology, or combination of technologies
- Don’t let the “best” or most familiar technology drive the model to build

- The place to start is asking questions:
  - What will this model be used for? Is it just stand alone analysis?
  - Who will this model be by/for? Management briefing or myself
  - Is this a one-time use, ongoing project, desktop tool for other workers?

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Data Analytics Stack
Tools: spreadsheets

- Excel is an amazing tool - “Jack of all trades master of none”
- The quantitative Swiss Army Knife for everyone and everything!
- “Spreadsheet Arthritis”
  - Because they are so useful, spreadsheets are used and continually added to, and therefore need to be maintained
  - The larger spreadsheets get, the more difficult they are to maintain
- For large data sets programmatic approaches are required for repetitive operations
- There are tremendous analytical tools available outside of Excel
- The real power is using the right tool, for the right task, and then integrating them together into a workflow

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Data Analytics Stack
Spreadsheets vs. python programming examples

When the problem involves...	Excel	Python
Small/moderate datasets	Excellent for small and moderate	Often unnecessary
Very large datasets	Scalability is challenging	Built for scale
Repetitive analysis	Power Query for data cleaning	Automate entire workflows
Complex transformations	Power Query can do a lot	More flexible and programmatic
Custom analytical logic	Formula/VBA limitations	Essentially unlimited logic
Statistical/ML methods	More limited	Huge ecosystem of libraries
Reproducibility	Possible, but spread across workbook	Code explicitly documents every step
Batch processing	Awkward	Natural - process many files like one
Interactive managerial model	Excellent	Usually not the best deliverable
Change assumptions in a meeting	Excellent	Often requires rerunning code

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File and Folder Structure

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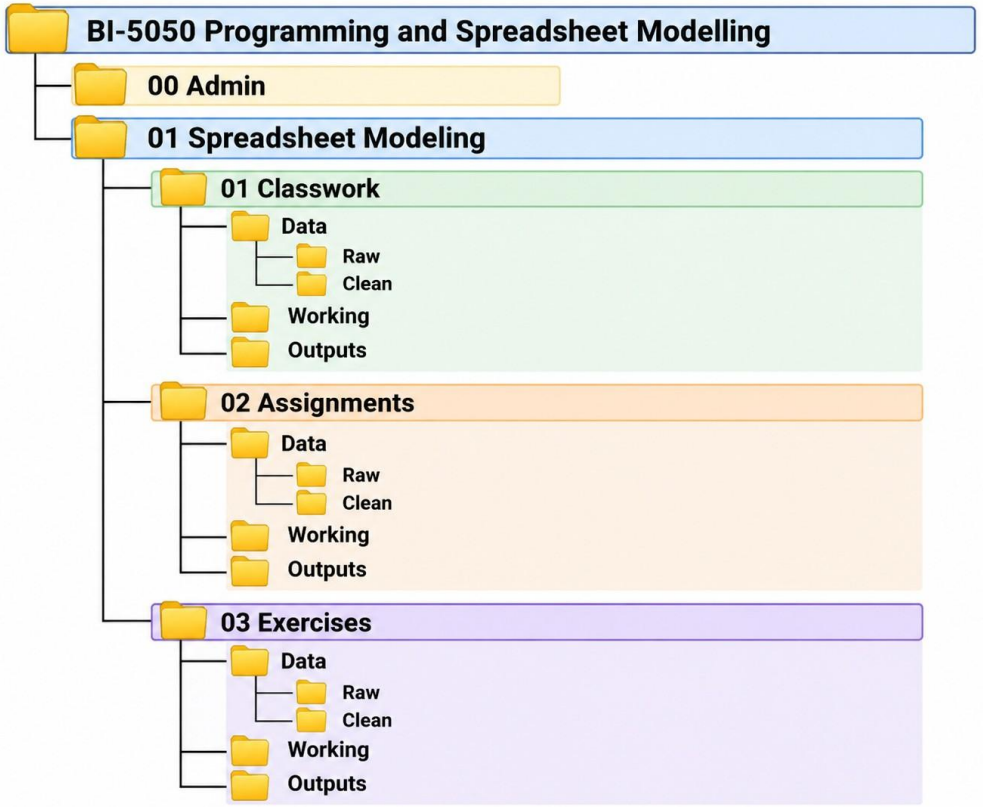
Folder Structure
Come up with what works for you!

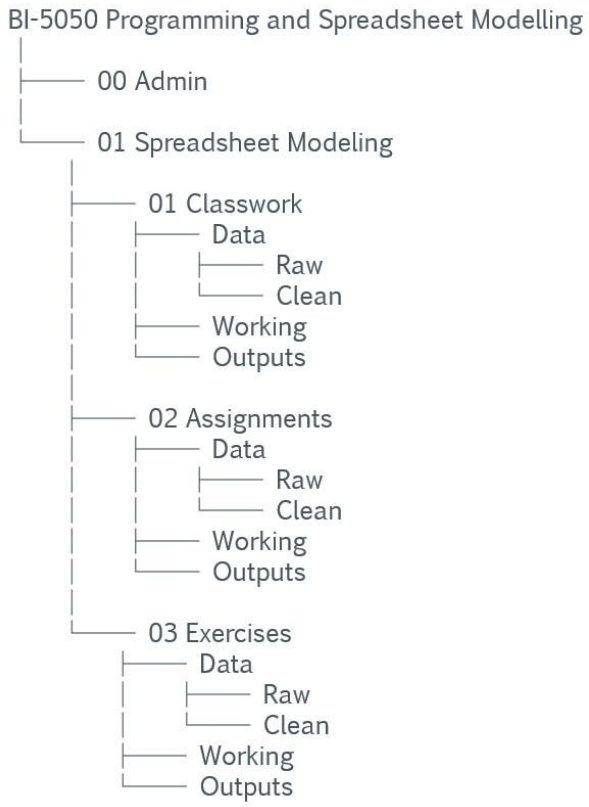
- You need a file organizational structure that works for you.
- Organization schemes are generally by:
  - Topic or week such as in school
  - Stages of development for projects
  - By data and products in analytics
  - Hybrid of one or more
- If you do not already have a preferred organization, one option might be a hybrid of weekly, which aligns with subject and data/products
- The greatest folder/file organization sucks if it doesn't work for **you**!
- The next page is just a suggested starting place.

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One potential folder/file organization

- These are the same – on left is ChatGPT image and on the right is what it looks like at a prompt
- If this isn't you, use AI and design yours from there





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